

Hull Printing Guide

BAMBU LAB X1C / P1S / A1 / A1 MINI · PLA · 0.4 MM NOZZLE

Printer routing

X1 Carbon, P1S, and A1: use the 220.1 × 177.3 mm `Hull.3mf` plate. A1 Mini: use the individual already-oriented STLs; all three hull modules are below 165 mm on their longest axis.

Required process

Setting	Hull modules	Running gear
Nozzle	0.4 mm	0.4 mm
Layer height	0.16 mm	0.12 mm
Walls	3	3
Top / bottom	5 / 5	5 / 5
Infill	15% gyroid	15% gyroid
Supports	Off	Off
Brim	Optional 3 mm	3 mm recommended

Orientation and support policy

The supplied manufacturing meshes are pre-oriented. Hull modules print top-interface down; shafts and A-brackets rest on hex facets; propellers and rudders lie flat. Disable supports and disable slicer auto-orientation. Internal key sockets bridge less than 8 mm.

Allowed color mapping

Objects	Approved filament
Hull / struts / rudders	PLA Matte Ash Gray
Shafts	PLA Silk Silver
Propellers	PLA Basic Gold

Fit and first layer

- Nominal concealed-joint clearance is 0.25 mm per side.
- Start with zero XY hole compensation. Tune only after a fit coupon.
- Use 0.10–0.15 mm elephant-foot compensation if needed; do not sand exterior seams.
- Do not independently scale STL files. Regenerate through the parameter source.

Preflight checklist

- Slicer reports 21 parts and a manifold model.
- Supports are off and orientations are unchanged.
- 0.12 mm object settings are applied to all 18 running-gear pieces.
- Shafts/struts have brim contact and no purge tower collision.

Post-print handling

Cool fully, flex the plate, and lift from robust hull surfaces. Never lever against the waterline groove, propeller blades, or rudders. Deburr only concealed interfaces.